Social Media Toxicity and Mental Health*

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Abstract

The impact of social media on users' well-being depends on both what they see and who they interact with. Even toxic content – harmful in many respects – may connect users to people and communities they value. This paper asks how reducing exposure to toxic content affects mental health. We randomly assign participants to a content-level intervention hiding toxic posts and comments on Facebook, X, and Reddit via a browser extension. We observe $N = 664$ desktop users over a six-week study period in two waves – one during the 2024 US election campaign and one outside of it. Hiding toxicity worsens mental health. The treatment increases an index of GAD-7 and PHQ-8 symptoms by 0.15σ, and the share of respondents with symptoms above the moderate cutoff rises by 12.1 pp. A QALY-based calibration indicates a welfare cost of about $158 per treated participant. We find evidence consistent with higher loneliness, lower relative moral self-view, and lower exposure to in-group political toxicity as mechanisms. We conclude that curbing toxic content can impose unintended psychological costs, even where it offers other social benefits.

JEL Classification: C93, D83, I10, I31, L82, L86, Z13

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1 Introduction

Social media platforms mediate a large share of information consumption and social interaction, and a growing body of causal evidence shows that they affect user well-being and mental health (Allcott et al., 2020; Braghieri et al., 2022; Allcott et al., 2022; Bursztyn et al., 2025; Allcott et al., 2025).[1] This work has focused predominantly on *access* – whether people use these platforms and how much. The content users are shown conditional on access is a distinct margin, which has received less attention, despite offering a more targeted policy lever: it can reduce harmful exposure while preserving the benefits of platform access.

Much of the intuition behind content regulation treats social media like traditional media: its effects on users operate predominantly through the content they view. By this logic, filtering toxicity, a commonly moderated type of harmful content, should improve mental health.[2] Moreover, a feed scrubbed of toxic content can leave users with a more benign picture of the world – perceiving discrimination and hostility as less pervasive – and such perceptions are linked to better mental health (Pascoe and Smart Richman, 2009). On the other hand, the platforms act as social networks: they shape the social environments in which users participate, not just the information they receive (Enikolopov et al., 2024; Donati and Song, 2026; Zhuravskaya et al., 2020). Toxicity is often woven into valued social participation. Even hostile exchanges signal that others are present and engaged, and removing this signal could increase loneliness.[3] Moreover, others' bad behavior provides a reference point against which users view themselves favorably. Overall, the media and social connections channels pull in opposite directions, leaving the net effect on mental health unclear.

We study this trade-off in a six-week field experiment on Facebook, X, and Reddit. A browser extension filtered toxic posts and comments for randomly assigned participants while preserving their access to the platforms. We find that filtering toxicity *worsens* mental health, increasing symptoms of both anxiety and depression. The mechanism evidence points to the social function of the platforms: filtering increases loneliness, and the adverse effects

[1] Studying mental health is critical given that mental health disorders are the leading cause of disability worldwide (Friedrich, 2017; OECD, 2021), affecting roughly one in two individuals over the life course (OECD, 2019). Moreover, poor mental health and psychological distress hinder human capital accumulation (Currie and Stabile, 2006, 2009; Eisenberg et al., 2009), contribute to the misallocation of talent (de Quidt and Haushofer, 2016), and are associated with penalties in the labor market (Biasi et al., 2021).

[2] Toxicity is defined as "a rude, disrespectful, or unreasonable comment that is somewhat likely to make you leave a discussion or give up on sharing your perspective". This definition applies to classifiers such as Perspective API and Unitary's Detoxify, deployed in both industry moderation systems and academic research (e.g., Kim et al., 2021; Jiménez Durán et al., 2025; Beknazar-Yuzbashev et al., 2025; Motte, 2026).

[3] Loneliness is a well-established determinant of mental health: it predicts the onset and persistence of depression and anxiety (Jin et al., 2026), and recent population evidence shows that it statistically mediates the link from anxiety and depressive symptoms to suicidal ideation (Musacchio Schafer et al., 2026). Thus, loneliness has become a first-order policy concern in its own right (Goldman et al., 2026).

are larger on platforms and in feeds where content is more strongly embedded in social ties and politically relevant communities.

We recruit desktop social media users via targeted ads on major platforms and through an online panel (CloudResearch) and ask them to install our extension. Participants are randomly assigned to (i) treatment conditions with toxicity hiding based on a Perspective toxicity score threshold of 0.33 or (ii) control conditions that hide nothing (pure control) or hide a matched random fraction of content to equalize hiding rates (active control).[4] The Perspective classifier maps text to a continuous toxicity score – the predicted fraction of human annotators who would label a snippet as toxic – and is built on human-labeled training data widely used in academic and platform settings (Wulczyn et al., 2017; Muralikumar et al., 2023). The study is conducted in two waves, one during the 2024 US election campaign and one outside the campaign period, and each wave runs for six weeks until a post-treatment survey. It then continues, with hiding still active, for an additional six weeks to a follow-up survey. Mental health outcomes (PHQ-8, GAD-7) are measured three times: in the pre-treatment, post-treatment, and follow-up surveys.

We compare changes from baseline between treated and control participants in a difference-in-differences design with respondent and cohort-by-time fixed effects. Our main result is that reducing exposure to toxic content worsens mental health. The index of mental health symptoms increases by 0.15σ (s.e. $= 0.05$, $p = 0.003$), and the anxiety and depression subindices increase by 0.13σ and 0.16σ respectively. On the raw clinical scales, the estimated increases are 0.7 for GAD-7 (on the 21-point scale) and 1.0 for PHQ-8 (rescaled to the full 27-point range). At the control arm means, these estimates imply that a 1% reduction in the share of toxic content corresponds to increases of 0.13% in GAD-7 scores and 0.15% in PHQ-8 scores.

The share of respondents with at least moderate symptoms on either scale (GAD-7 or rescaled PHQ-8) increases by 12.1 percentage points ($p < 0.001$). Using the more conservative PHQ-8 ≥ 10 cutoff, the effect is 9.3 percentage points ($p = 0.006$). Considering individual symptoms, we find worsening across GAD-7 items. For PHQ-8, the effects are more concentrated, in sleep disturbance, depressed mood, and negative self-evaluation. A conservative QALY-based calibration, which values lost quality of life at respondents' own reported household income, indicates a welfare cost of about \$158 per treated participant over the six weeks.

We benchmark our effects against two RCTs that measure mental health on clinical scales.

[4]Participants in the treatment conditions were randomly assigned to one of two variants: one in which toxicity hiding was solely based on whether the Perspective score exceeded 0.33 and one that additionally screened content when the Alienation score exceeded 0.7. See Section 2.2 for more details.

Angelucci et al. (2026) document a 0.29σ improvement in a composite PHQ-8/GAD-7 index at six months from an AI-powered cognitive behavioral therapy app among distressed women. Shreekumar and Vautrey (2024) find that randomized access to the Headspace mindfulness app reduces a composite anxiety-depression index by 0.46σ after four weeks. Our 0.15σ effect is in the opposite direction and equal to about 52% of the Angelucci et al. benefit, measured over less than a quarter of their follow-up period, and 33% of the Headspace improvement.[5]

Our interpretation of the results stems from the distinction between consuming content and maintaining social connections through a platform. Users may still value a toxic post if it lets them hear from friends or see their political views represented. Filtering these posts reduces exposure to toxicity but may also take away some social benefits of platform use. We use secondary survey outcomes and heterogeneity in the ITT to examine this possibility.

Among the secondary survey outcomes, the clearest response to toxicity filtering is an increase in loneliness, by 0.34σ ($p = 0.003$). This estimate remains statistically significant after correcting for multiple testing across secondary outcomes. We find no detectable changes in anger, fear of missing out, or perceived discrimination.[6]

The adverse mental-health ITT is larger among users with a higher share of feed impressions from accounts classified as peers. Conditional on platform use shares, a one-standard-deviation increase in this share is associated with a 0.11σ larger ITT ($p = 0.013$).[7] Moreover, the implied platform-specific ITTs are 0.17σ for Facebook and 0.21σ for X – platforms organized around interpersonal exchange – compared with 0.01σ for Reddit.

Political affiliation provides a second way to examine the relationship between users and the content being filtered. The adverse ITT is larger in magnitude for Democrats and Republicans than for independents. Among toxic political posts with a discernible partisan affiliation, most favor the user's own party. Users who view the opposing party as a threat may value posts showing that others share their concerns. Consistent with this interpretation, the adverse ITT is larger among respondents with higher pre-treatment hostility toward the opposing party, measured by the moral disengagement index (MDI): a

[5]Two further RCTs offer useful benchmarks. Bhat et al. (2022) report depression reductions from two behavioral activation psychotherapy trials in India four to five years on, of up to 0.23σ for the more effective Healthy Activity Program, which reduced the incidence of moderate depression by 13 pp. Our adverse effect is of similar size: the share of respondents above the moderate-symptom threshold on either scale rises by 12.1 pp. Francois et al. (2025) conduct the largest psychedelics RCT to date and find that a one-off ayahuasca treatment raises happiness and lowers distress by roughly 0.4σ at six months among 429 first-time users. Our effect is about 38% of that benefit, with the sign reversed.

[6]Respondents also assess their own positive qualities less favorably relative to the average person (-0.19σ, $p = 0.037$). Seeing others behave badly may make respondents judge themselves more favorably. The coefficient is no longer statistically significant after correction for multiple testing across secondary outcomes.

[7]These accounts are identified using their reach, engagement, and verification status, so the measure proxies for peer content without directly measuring personal relationships.

one-standard-deviation increase in MDI is associated with a 0.12σ larger ITT ($p = 0.006$). The peer-share and MDI coefficients remain positive when entered jointly.

It is important to interpret our content-level results in light of the literature on social media access and mental health. A set of randomized experiments, spanning platform deactivations and shorter abstinence periods, consistently finds that quitting social media modestly improves subjective well-being (Tromholt, 2016; Turel et al., 2018; Hall et al., 2019; Lambert et al., 2022; Arceneaux et al., 2024). Allcott et al. (2020) find that deactivating Facebook for four weeks before the 2018 US midterms improved subjective well-being by 0.09σ, showing harmful effects of platform access. Relatedly, Allcott et al. (2025) find that deactivating Facebook before the 2020 US election improved an index of happiness, depression, and anxiety by roughly 0.06σ. Braghieri et al. (2022) report that Facebook's rollout across US college campuses (2004–2006) worsened mental health, measured by NCHA instruments, by 0.085σ.

Our sizable 0.15σ effect of toxicity hiding on mental health is in the opposite direction, highlighting the importance of understanding access effects as *net* effects of various channels. These earlier findings are consistent with ours: for example, unfavorable social comparisons, a mechanism proposed by Braghieri et al. (2022), may harm mental health, while social connections may improve it. Deactivation reduces exposure to unfavorable comparisons but can also disrupt important social connections, so its modest net benefits could reflect substantial gains partly offset by a connection cost. Reducing social media use while preserving social connections could therefore yield larger mental-health gains, and testing this is an avenue for future research. Under a simple additive decomposition, these gains could be around 0.24σ, close to the 0.29σ improvement from an AI-delivered CBT app (Angelucci et al., 2026).

Given the potentially addictive nature of social media, an alternative interpretation is that removing toxicity induces temporary withdrawal, with mental health initially deteriorating before improving. We find no support for it. The time scale of this study extends beyond the typical acute withdrawal periods for addictive goods such as caffeine or nicotine, and evidence on smoking cessation shows positive effects already at comparable horizons.[8]

To strengthen the interpretation of our main results, we conduct a battery of robustness checks. The effect on the index of mental health symptoms is stable across time horizon (adding the twelve-week follow-up), compliance window (restricting to respondents active for at least one or three weeks), recruitment channel (CloudResearch vs. social-media ads), and classifier (toxicity-only versus alienation-augmented arm). Our results replicate across two recruitment waves, one during an election campaign and one outside of it, suggesting the estimate is unlikely to be an artifact of an election period. We find no evidence of differential

[8]Furthermore, withdrawal tends to be shorter and milder when dependence is weaker. See Section 6.2 for a full discussion of withdrawal as a potential alternative explanation of the results.

attrition by treatment status – treatment assignment does not predict post-treatment survey completion or continued extension activity across any specification. The results are also robust to reweighting for survey non-response on observables.

Our design allows us to rule out generic feed disruption as an explanation: the random-hiding arm, which removes a matched fraction of non-toxic content, produces no statistically detectable effect on mental health, while the effect of hiding toxicity remains positive when the random-hiding group serves as the comparison, with a coefficient of 0.20σ ($p = 0.002$). The adverse ITT estimated at the six-week post-treatment survey persists over time, with the symptom index remaining elevated at the twelve-week follow-up, which is inconsistent with a purely transitory adjustment response. Treated respondents were not more likely to report changes in their social media experience than control respondents. This argues against explanations based on experimenter demand or a perceived loss of control over their feeds. Moreover, we find no indication of substitution toward mobile usage and provide evidence that the results are unlikely to be explained by substitution toward websites more toxic than the treated platforms. We also validate our hiding decision rule with two external classifiers and provide evidence against sizable classification error. Lastly, we show that the treatment has no effect on the topical composition of the feed across 25 categories, with effects on the toxic share of content being an order of magnitude larger than for the topic shares.

We assess the representativeness of our sample in two ways. First, its demographic composition is broadly in line with platform benchmarks on age, gender, race, and college education, though it leans more Democratic and includes users with higher desktop shares of social media consumption. Second, we benchmark baseline mental health against a supplementary sample of US social media users on Prolific, reweighted to match platform demographics. Average GAD-7 and PHQ-8 scores are 5.59 and 6.81 in this sample, compared with 5.21 and 5.85 in ours, indicating similar and, if anything, slightly lower symptom levels. While our sample is not fully representative, its participants do not have unusually poor mental health relative to typical social media users.

Our findings do not imply that content moderation is futile or harmful. The harms of toxic content are well-established: hate speech online has been linked to a global increase in violence toward minorities (Council on Foreign Relations, 2019; Müller and Schwarz, 2021), and reducing the supply of hateful content can curb offline hate crimes (Jiménez Durán et al., 2025). Rather, our results call for a more granular, user-centered approach to how moderation is implemented. One alternative is friction-based flagging: platforms could label posts that exceed a certain toxicity threshold and let users decide whether to view them. The larger adverse ITT among users whose feeds contain more peer content also suggests testing lighter restrictions on posts from peers than on those from influencers or institutions.

We contribute to the literature along four dimensions. First, we isolate a distinct content-level channel – exposure to toxic material – within the broader effects of social media on mental health and well-being. Unlike designs that jointly alter time online, content exposure, and social network exposure (Allcott et al., 2020; Braghieri et al., 2022; Allcott et al., 2025; Mosquera et al., 2020; Asimovic et al., 2021; Collis and Eggers, 2022) or study the effects of algorithm personalization (Levy, 2021; Guess et al., 2023a; Mandile, 2025; Gauthier et al., 2026), our intervention directly manipulates toxicity exposure while holding platform access and other features of the recommendation algorithm constant.[9] Our finding that filtering toxicity worsens rather than improves mental health shows that, for the viewing user, exposure to toxic content is not a net driver of the adverse mental-health effects of social media: hiding it does not alleviate them and can itself impose a cost. This suggests that care should be taken in designing content moderation policies, as seemingly beneficial interventions may have unintended psychological costs.

Second, we contribute to two related literatures: on the effects of harmful content and its moderation, and on the effects of recommendation algorithms. We extend research on toxic content – which is more prevalent following online disputes (Meng et al., 2026), affects user engagement (Beknazar-Yuzbashev et al., 2025, 2024), and, when weaponized by politicians, voters' intentions (Motte, 2026) – and on the content moderation policies designed to curb it (Andres and Slivko, 2021; Jiménez-Durán, 2023; Ahmad et al., 2024; Rizzi, 2024; Kominers and Shapiro, 2024; Madio and Quinn, 2024; Di Liza and Shevchenko, 2026; Jiménez Durán et al., 2025). We do so by documenting that such content, and its removal, affect clinically validated depression and anxiety outcomes. More broadly, we add to a growing body of work on recommendation algorithms, which has studied the effects of overall personalization (e.g., Gauthier et al., 2026) and of specific alternative algorithms (Nyhan et al., 2023; Guess et al., 2023b; Piccardi et al., 2025; Milli et al., 2025; Allcott et al., 2026; Stray et al., 2026).

Third, we provide new evidence on the mechanisms linking content environments to mental health, identifying social connection – through loneliness (Enikolopov et al., 2024; Musacchio Schafer et al., 2026; Jin et al., 2026), horizontal ties, and political attachment – as the operative channel. Through these findings, we connect the literature on determinants of mental health to studies on the psychology of partisan identity (Graham et al., 2009; Rathje et al., 2021; Brady et al., 2017), to work on how others' comments shape engagement and attitudes (Donati and Song, 2026), and to the literature on how network structure influences

[9]This literature consistently finds that social media access worsens subjective well-being and mental health. Aridor et al. (2024) offer a useful review. Evidence on other media technologies is more mixed. Exploiting quasi-random variation in commercial television reception in West Germany, Chadi and Hoffmann (2026) find no adverse health effects and, if anything, positive effects on life satisfaction. Golin (2022) finds that broadband Internet access in Germany worsened mental health among women but not among men.

platform welfare effects (Burke and Kraut, 2016; Valkenburg and Peter, 2009).

Fourth, methodologically, we extend the growing toolkit of browser-extension experiments that directly manipulate on-platform content without platform collaboration (Aslett et al., 2022; Yu et al., 2024; Allcott et al., 2024; Farronato et al., 2024; Beknazar-Yuzbashev et al., 2025; Piccardi et al., 2025; Stray et al., 2026).

The paper is structured as follows. Section 2 describes the experimental design. Section 3 presents the data, main outcome measures, and empirical framework. Section 4 reports the effects on mental health. Section 5 presents evidence on mechanisms. Section 6 examines robustness and alternative interpretations. Section 7 concludes.

2 Experimental Design

2.1 Intervention and Implementation

We used a Chromium-based desktop browser extension to introduce exogenous variation in exposure to toxic content on three social media platforms – Facebook, X, and Reddit – while leaving platform access otherwise unchanged.[10] The extension achieves that by modifying the page's Document Object Model (DOM) to seamlessly hide targeted posts and comments. Figure 1 shows a toxic Facebook post being hidden, with the content below shifting up to close the gap. An analogous process was applied to tweets on X and posts on Reddit.

Comments are hidden in the same way on X and Reddit, whereas on Facebook a hidden comment is instead replaced with the italicized placeholder "Deleted comment."[11] Two platform-specific details reflect how each site structures conversations. On Reddit, replies are nested beneath the comment they answer, so hiding a comment also hides its replies. On X, where a single post can span several linked tweets (a "thread"), the original poster's own tweets are exempt within a conversation, so that such a thread is not fragmented; replies left by other users are hidden when toxic. The participant's own posts and comments are never hidden, regardless of their toxicity.

Hiding is fast: in Beknazar-Yuzbashev et al. (2025), which deploys a similar browser extension, the median hiding time is about 400 milliseconds. The extension classifies multiple items at once, and platforms load most content in batches before it comes into view. At this speed, flagged items are usually hidden before users have the possibility to see them, ren-

[10] Chromium-based browsers (e.g., Chrome, Edge, Opera) cover 87% of desktop browsing market share as of May 2026; https://gs.statcounter.com/browser-market-share/desktop/worldwide, accessed: 2026-06-06.

[11] At the time of the experiment, the technology that hid Facebook comments was unstable in testing, and thus we opted for a more reliable moderation tool. Our results are robust to using the random-hiding control, in which the prevalence of Facebook moderation via "Deleted comment" is approximately the same as in the treatment group.

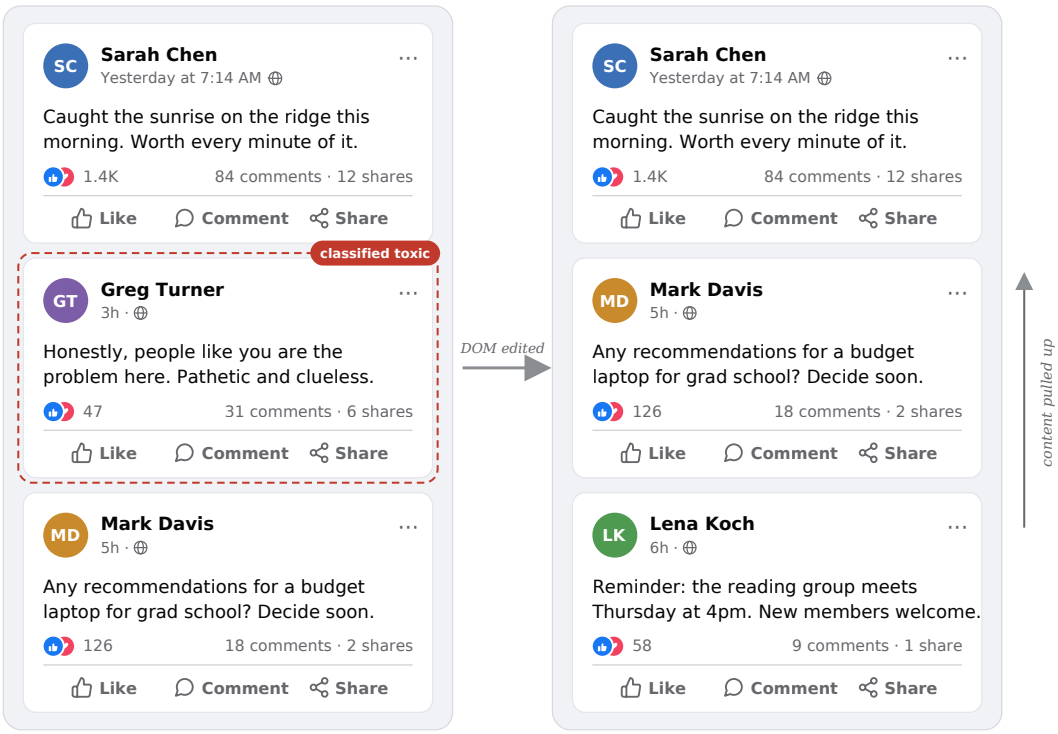


FIGURE 1: HOW THE EXTENSION HIDES TOXIC POSTS

Notes: The figure illustrates the post-hiding mechanism on a stylized Facebook feed. *Left:* the feed as it is intended to be delivered by the platform, with the post flagged as toxic by the extension's classifier outlined in red and labeled here for exposition only. *Right:* feed as the user sees it: the container of the post removed from the page's DOM, and the posts below shift up to fill the vacated space, so that no gap or placeholder remains visible to the participant. The same procedure applies to posts on X and Reddit. The names and texts presented here are fictional.

dering their experience effectively seamless. A dedicated follow-up experiment there showed that neither longer hiding times nor the hiding itself affects user experience. On X and Reddit, where feasible, the present extension goes further: newly loaded content is held invisible for a brief interval (to allow the hiding decision to finalize), after which it appears only if it was not flagged.[12] Despite these efforts, we do not rely on speed or seamlessness for identification. Instead, we hold user experience constant through a random-hiding control condition, described in Section 2.2.

[12] As content loads, the extension immediately makes each new post or comment invisible – while preserving the space it occupies, so the page does not jump – and the toxicity classifier then evaluates it; the item is revealed if it is not flagged and kept hidden if it is. This was not feasible on Facebook.

2.2 Treatment Arms and Randomization

After installing the extension, participants were randomized into treatment (60%) or control (40%), and evenly into two conditions within each arm (see Figure 2).[13] All conditions score text content with Jigsaw's Perspective API, which assigns each post or comment a toxicity score equal to the predicted share of human annotators who would rate it toxic – that is, "a rude, disrespectful, or unreasonable comment that is somewhat likely to make you leave a discussion or give up on sharing your perspective". Our threshold of 0.33 thus flags content that at least a third of annotators would label toxic. The two treatment conditions differ only in the hiding rule: one hides any content above this threshold; the other additionally hides content scoring above 0.7 on Alienation, an experimental attribute of the Perspective API.[14]

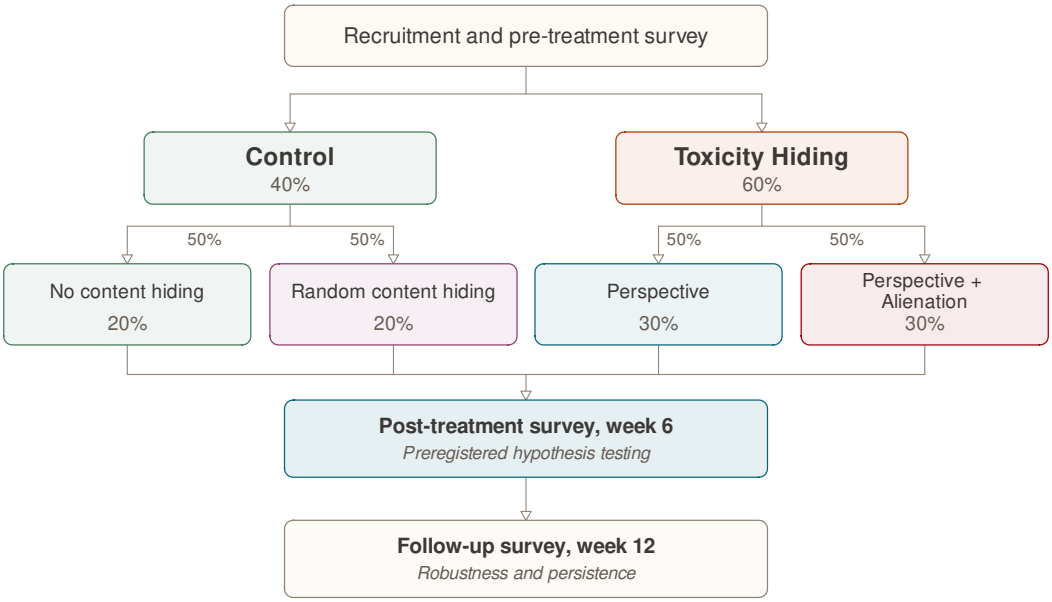


FIGURE 2: EXPERIMENTAL FLOW

The choice of the threshold reflects the study's focus on the everyday social environment users encounter online. An online environment may feel hostile and unwelcoming even without vile hate or severe harassment (usually considered to score above 0.7 on the toxicity scale

[13] For Wave 1, the initial assignment probabilities into treatment and control were flipped relative to what was preregistered. As a result, we oversampled the control arm. On August 2, 2024 we re-randomized a randomly selected subsample of users assigned to the control arm into one of the treatment conditions to match the targeted assignment probabilities. Results are robust to dropping participants who had been initially assigned to the random-control arm and were then re-randomized into treatment (Online Appendix).

[14] Alienation is a Perspective attribute developed by Google Jigsaw. Schmer-Galunder et al. (2024), who benchmark it among six prosocial dimensions, define it as "language that refers to an opposing individual or group as less human, inferior, and/or not fitting in within the norms of a social group or society."

and often quickly moderated by platforms). 'Ordinary' incivility, disparaging remarks, and antagonistic exchanges can be unpleasant and disturbing for a substantial subset of readers, while the model would predict less agreement about their toxicity. These are precisely the interactions that would not be filtered under a higher cutoff choice.[15]

The two controls differ in whether anything is hidden. The no-hiding (pure) control hides nothing, only recording and scoring content. The random-hiding (active) control hides content at random, rather than by toxicity, with the rate calibrated to approximate the treatment arms. Comparing treatment with random hiding isolates the effect of hiding toxic content, holding constant both the amount hidden and any possible effects of the hiding process itself.

Our headline comparison pools the two treatment conditions and the two control conditions, contrasting any toxicity hiding with either control. We preregistered this pooling, and the data support it: both the random-versus-no-hiding difference and the difference between the two treatment rules are null – see Section 6. In the same section, we report the variant-specific comparisons for robustness.

2.3 Study Flow

We recruited participants through ads on Facebook and X and through the CloudResearch panel, restricting eligibility to English-speaking U.S. residents on a Google Chrome desktop browser, the environment in which the extension operates. To limit selection into the study on toxicity preferences, recruitment materials described the study only as a paid opportunity to improve one's social media experience and did not disclose the extension's functionality (see examples of recruitment ads in the Online Appendix).

Figure 2 summarizes the study flow. After installing the extension, participants completed the pre-treatment survey and were randomized to experimental conditions. Six weeks later they completed the post-treatment survey, our primary preregistered measurement, which we compare against the pre-treatment baseline.[16] To assess whether the effect persists, we re-surveyed participants at twelve weeks (the follow-up survey). Due to higher expected attrition, we intended it as a suggestive robustness check. Participants were remunerated through a mixture of draws for Amazon gift cards of up to $500 and flat payments

[15] See Section 6.2 for a discussion of validation of our classifier and the hiding rule using two external classifiers of incivility and harassment/hate. We show that considerable shares of content flagged by the external classifiers have Perspective scores in the range of $[0.33, 0.7]$.

[16] Some users were recruited before the intervention functionality was ready for production, and they faced a longer period between the pre-treatment survey and the treatment start. The median difference is 1 day.

for survey completion.[17] We ran two waves of identical design – the first from July 16 to November 5, 2024, spanning the U.S. presidential election campaign, and the second from May 8 to September 6, 2025, outside any campaign – to test whether effects are specific to a highly politicized period. In total, 1394 participants enrolled.

This experiment (preregistered as AEARCTR-0013987) was one component of a larger research platform: the same extension, randomization, recruitment, and questionnaire simultaneously supported a separately preregistered political attitudes study (AEARCTR-0013997) conducted by a partially overlapping but distinct set of investigators. The questionnaire included a block of mental health questions, analyzed here, and a block of political attitudes questions, analyzed in the other paper. The two studies ex ante preregistered separate sets of families of primary outcomes – clinical symptom scales for this paper and political attitudes for the other – and were planned as separate publications.[18] Each registry entry cross-references the other study. We use political attitudes only to examine heterogeneity in the mental-health ITT and restrict them to pre-treatment outcomes. All arms received both blocks at each of the three surveys, so questionnaire length did not vary with treatment assignment. Any common influence of survey length on reported symptoms cancels out when we compare changes across arms.

2.4 Sample Definition

Our starting sample comprises the 1394 participants who completed the pre-treatment survey and installed the extension. We compare the four arms on fourteen pre-treatment characteristics: age, sex, race, education, party affiliation, recruitment channel, media-use intensity, desktop shares for each platform, and anxiety and depression. Our arms are well-balanced, and an F-test of equality across the four arms fails to reject at the 5% level, as expected under random assignment (balance table in the Online Appendix).

For the main analysis, we retain participants who completed the post-treatment survey and recorded extension activity on a supported platform during the study, yielding $N = 664$. Attrition in this sample is unrelated to treatment, and we probe sensitivity to alternative activity requirements in Section 6.

[17] After installing the extension and completing the pre-treatment survey, all participants were entered into a draw for Amazon gift cards of up to $500. In addition, participants recruited via social media ads were paid $10 for completing both the post-treatment and follow-up surveys, while participants recruited via CloudResearch were paid $5 for each of the pre-treatment, post-treatment, and follow-up surveys.

[18] The separately preregistered political attitudes study has 4 primary outcomes that are outside this paper's estimand. Section 6.1 shows that our main inference is robust to correcting for multiple hypothesis testing, including in families that add these outcomes.

2.5 Design Tradeoffs

Our design has no pre-treatment baseline period for extension activity – the intervention begins as soon as participants complete the baseline survey and are randomized. Collecting pre-treatment activity data would have added weeks to the study and increased the (expected) attrition. Starting treatment immediately kept the study short and preserved power for the mental-health ITT. Without those data, however, we cannot compare participants' engagement before and after treatment and have limited power to estimate the engagement ITT. We therefore do not study engagement here; its response to hiding toxic content is already documented in Beknazar-Yuzbashev et al. (2025), whose design supports such within-person analysis.

Second, because the extension operates only on desktop devices, we recruited users who browse social media primarily there, which maximizes the share of their consumption it can screen. The resulting sample is a selected segment of the population of social media users. We examine its representativeness in Section 3.2.

3 Data, Outcomes, and Estimands

This section discusses the data, mental-health outcome measures, and empirical framework. Secondary outcomes and heterogeneity angles used for mechanism analysis are deferred to Section 5.1. Our approach is based on our preregistration and any deviations or extensions are disclosed in a dedicated section of the Online Appendix.

3.1 Primary Outcomes

The primary outcome is based on two psychometric scales widely used in clinical practice to screen for anxiety and depressive disorders: the Generalized Anxiety Disorder scale 7 (GAD-7) and the Patient Health Questionnaire 8 (PHQ-8) (Spitzer et al., 2006; Kroenke et al., 2009). Each instrument asks respondents to indicate, on a four-point scale from "not at all" to "nearly every day," the frequency with which they experienced a set of symptoms over the preceding two weeks.[19] Responses are summed to yield a non-negative integer score: GAD-7 ranges from 0 to 21 and PHQ-8 from 0 to 24. Higher values denote higher incidence of mental health symptoms and therefore worse mental health.

We follow the recognized practice of using PHQ-8 instead of PHQ-9 (Centers for Disease Control and Prevention, 2011), because the ninth item, concerning thoughts of death or self-harm, can require clarification and further clinical assessment that would be difficult to

[19] The exact wording is provided in the Online Appendix.

provide promptly in our online survey setting. Omitting this item preserves most of the instrument's performance, as PHQ-8 and PHQ-9 scores are highly correlated ($r = 0.996$; Wu et al., 2020). Following prior applications of proportional rescaling (Syrjala et al., 2022; Hirshberg et al., 2024), we additionally construct a *PHQ-9 proxy* by rescaling the PHQ-8 sum by 9/8, placing the depression measure on the same 0–27 range as PHQ-9. With integer PHQ-8 scores, the standard moderate-symptom cutoff of PHQ-9 $\geq$ 10 is equivalent to PHQ-8 $\geq$ 9.

Our headline outcome is a standardized composite index of mental health symptoms, defined as the simple average of the z-scorized PHQ-8 and GAD-7 items and normalized to mean zero and unit standard deviation using the control pre-treatment distribution. All primary outcomes are elicited in each of the three surveys (pre-treatment, post-treatment, and follow-up).

3.2 Descriptive Statistics

We summarize observable characteristics of the main sample of $N = 664$ participants used for the headline six-week ITT estimates and compare them against the population of US social media users. Table 1 provides the full descriptive statistics.

Respondents' age, gender, race, and college education broadly align with values spanned by Facebook, Reddit, and X users. Our sample is skewed toward Democrats and individuals with lower household incomes, though we retain considerable shares of Republicans (20.5%) and individuals with household incomes above $100,000 (23.9%). As a result of our recruitment strategy, which is necessary to ensure sufficient intervention strength, our sample consumes social media primarily on desktop – the reported desktop shares of consumption are 61.0% for Facebook, 66.7% for Reddit, and 60.8% for X. While a typical social media user consumes the treated platforms relatively more on mobile devices, their desktop shares remain considerable – the lowest is on Facebook at about 35–40%.

To our knowledge, no other dataset documents PHQ-8 and GAD-7 scores for a population of social media users, so we benchmark the pre-treatment scores in our sample in two ways. First, we compare our sample to the set of all CloudResearch respondents who finished the pre-treatment survey, regardless of completing the post-treatment survey or activating the extension (i.e., our main sample includes a subset of such CloudResearch respondents).[20] Second, in October 2026 we recruited a supplementary sample of 100 US social media users

[20]The Online Appendix offers further comparison between participants recruited through ads vs. CloudResearch, as well as with a nationally representative sample of US adults.

on Prolific.[21] Reweighting the observations to match platform demographic benchmarks, we find mean GAD-7 and PHQ-8 scores of 5.59 and 6.81. These are similar to our field sample, where the means are 5.21 and 5.85, respectively.

Taken together, these comparisons show that our sample is not fully representative of US social media users. At the same time, it is unlikely that our results can be explained by recruiting a sample of individuals with particularly poor mental health relative to typical social media users.

TABLE 1: DESCRIPTIVE STATISTICS

	Study sample		Benchmark					
							Prolific	
Variable	Mean / Share	N	Facebook	Reddit	X	CR	Unweighted	Reweighted
Panel A. Demographics (vs. platform users)								
Age (years)	41.17	664	47	38	41	38.70	42.63	43.79
Male	45.6%	664	44%	55%	60%	51.3%	49.0%	49.6%
White	67.0%	664	60%	64%	53%	64.7%	64.3%	59.7%
Bachelor's degree or higher	52.0%	663	39%	53%	44%	52.8%	65.0%	43.0%
Income $\geq$ \$100k	23.9%	664	30%	42%	36%	26.6%	34.0%	33.6%
Panel B. Political affiliation (vs. platform users)								
Democrat	47.9%	664				48.2%		
Republican	20.5%	664				22.2%		
Independent	25.3%	664				24.7%		
Unaffiliated / no answer	6.3%	664				4.8%		
Democratic (incl. leaners)	69.5%	663	50%	60%	45%	67.8%		
Panel C. Recruitment and baseline platform use (vs. site traffic)								
Cloud Research sample	66.3%	664						
Self-reported social-media use (min/day)	116.49	663				109.24		
Share of Facebook use on desktop	61.0%	597	~35–40%			63.2%		
Share of Reddit use on desktop	66.7%	552		~55–60%		69.7%		
Share of X use on desktop	60.8%	479			~40–45%	62.9%		
Panel D. Baseline mental health (vs. Cloud Research and Prolific users)								
Baseline PHQ-8 score	5.85	664				5.77	6.90	6.81
Baseline PHQ-9 proxy score (PHQ-8 x 9/8)	6.58	664				6.50	7.76	7.66
Baseline GAD-7 score	5.21	664				5.29	5.68	5.59
Baseline PHQ-8 $\geq$ 10	23.6%	664				23.4%	37.0%	37.0%
Baseline PHQ-9 proxy $\geq$ 10	27.4%	664				27.3%	38.0%	38.3%
Baseline GAD-7 $\geq$ 10	19.1%	664				19.2%	26.0%	25.5%
Either PHQ-9 proxy or GAD-7 $\geq$ 10	30.9%	664				31.1%	43.0%	41.7%

Notes: The study sample is the main sample ($N = 664$). The demographic benchmarks for Facebook, Reddit, and X estimate the composition of each platform's US adult user base. We calculate them using Bayes' rule, combining platform-use rates within demographic categories from Pew Research Center's (2025) with those categories' shares of the US adult population. Mean age is approximated using age-band midpoints. CR denotes the CloudResearch reference sample ($N = 598$). It includes all participants recruited through the panel who completed the pre-treatment survey, regardless of whether they qualified for the main sample and whether they completed the post-treatment survey. The limitation of that sample is that it represents CloudResearch panelists selected on their agreement to install a browser extension. Prolific denotes a supplementary sample of 100 users of Facebook, X, or Reddit recruited on Prolific in October 2026. The Unweighted column reports participants as recruited, and the Reweighted column reweights them to the average platform user on observable demographics. The entropy-balancing weights (Hainmueller, 2012) match the five Panel A characteristics exactly to the Facebook, Reddit, and X user bases, pooled in proportion to each platform's number of US adult users. Age is matched on the same band midpoints as the platform columns. The effective sample size under the weights is 77.

[21] We screened participants on the following answers to the question "Which of the following social media sites do you use on a regular basis (at least once a month)?": Facebook, X, or Reddit.

3.3 Empirical Framework

Setup. We consider individuals i and periods $t \in \{0, 1\}$, where 0 corresponds to the pre-treatment survey and 1 to the six-week post-treatment survey. $D_i \in \{0, 1\}$ denotes assignment to the pooled treatment arm (toxicity hiding). $\text{Treated}_{it} \equiv D_i \cdot \mathbf{1}\{t = 1\}$ is an indicator equal to one for treated individuals in the post-treatment period and zero otherwise. Each individual has an intervention start date $s(i)$ – users with the same start date form a start-date cohort. We also define an individual fixed effect α_i and a cohort-by-time fixed effect $\gamma_{t \times s(i)}$. All cohorts fall entirely within one of the two recruitment waves, so cohort-by-time fixed effects absorb any wave-specific time shocks. Y_{it} is an outcome variable, such as the standardized index of mental health symptoms.

We estimate the intent-to-treat (ITT) effect of assignment to toxicity hiding. Because treatment is randomly assigned, parallel trends holds in expectation, and our sample is balanced on observables. Individuals enrolled and began treatment on different days, so we condition on the start-date cohort through $\gamma_{t \times s(i)}$, which absorbs cohort-specific time effects. As a result, our identification relies only on comparisons of treated users against never-treated users in the same cohort and period. We cluster the standard errors at the individual level i.

Main specification. Our headline specification is

$$Y_{it} = \beta \, \text{Treated}_{it} + \alpha_i + \gamma_{t \times s(i)} + \varepsilon_{it}. \tag{1}$$

β is the coefficient of interest and it measures the ITT of assignment to toxicity hiding on outcome Y_{it}. Following our preregistration, we pool the two treatment conditions and the two control conditions and conduct a single treated-vs-control analysis. We report the variant-specific comparisons in Section 6.

Exposure response. To understand the impact of a change in the share of toxic content, we instrument the realized share E_{it} with the treatment assignment indicator:

$$Y_{it} = \pi \, \widehat{E}_{it} + \alpha_i + \gamma_{t \times s(i)} + \varepsilon_{it}, \qquad E_{it} = \rho \, \text{Treated}_{it} + \alpha_i + \gamma_{t \times s(i)} + u_{it}. \tag{2}$$

Since we do not have a baseline period, we set $E_{i0} = 0$ for all participants (this common shift is absorbed by the cohort-by-time fixed effects). Thus, we interpret ρ as the difference in the share of toxic content between the treatment and the control group during the six-week intervention period. Under the assumption that treatment assignment affects the outcome Y_{it}

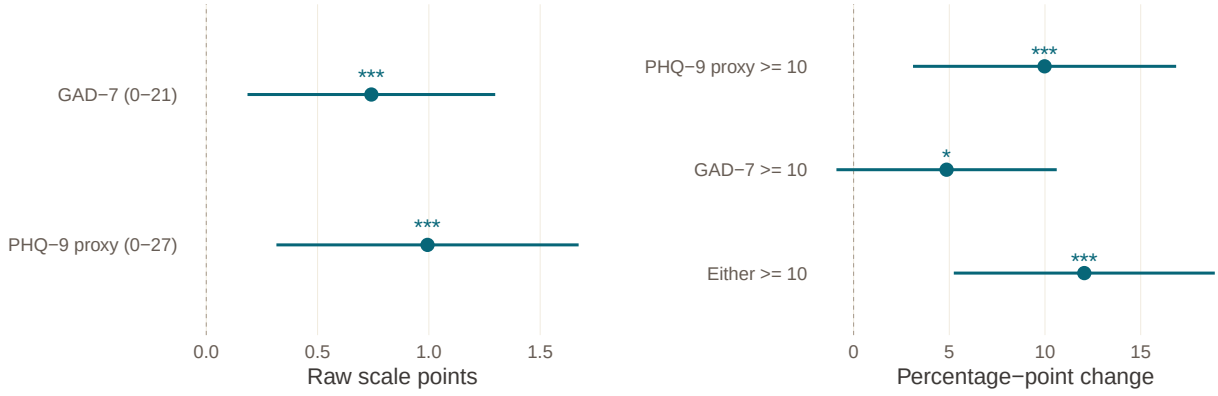
only through exposure to toxicity, π represents an exposure-scaled ITT effect. In Section 6, we find no statistically significant effect of the random-hiding control relative to the pure control, which is consistent with a valid exclusion restriction. We note that, since E_{it} is continuous, π is a dose-response coefficient.

4 Main Results

4.1 Effects on Mental Health

Clinical scales. Toxicity filtering worsens mental health. The index of mental health symptoms rises by 0.15σ ($p = 0.003$). The anxiety and depression subindices both increase, with coefficients of 0.13σ and 0.16σ, respectively. The same pattern is visible on the symptom scales: GAD-7 rises by 0.7 points on the 21-point scale and the PHQ-9 proxy rises by 1.0 points on the 27-point scale (Figure 3, left panel).

Threshold crossings. Toxicity filtering also shifts respondents across conventional cutoffs for moderate symptoms (Figure 3, right panel). The share of individuals scoring at or above the PHQ-9 proxy cutoff of 10 rises by 10.0 pp ($p = 0.005$) and the share scoring at or above the GAD-7 cutoff rises by 4.9 pp ($p = 0.098$). The share breaching either cutoff rises by 12.1 pp ($p < 0.001$). The result is robust to using the more conservative PHQ-8 ≥ 10 rule, with the share breaching either cutoff rising by 9.3 pp ($p = 0.006$).



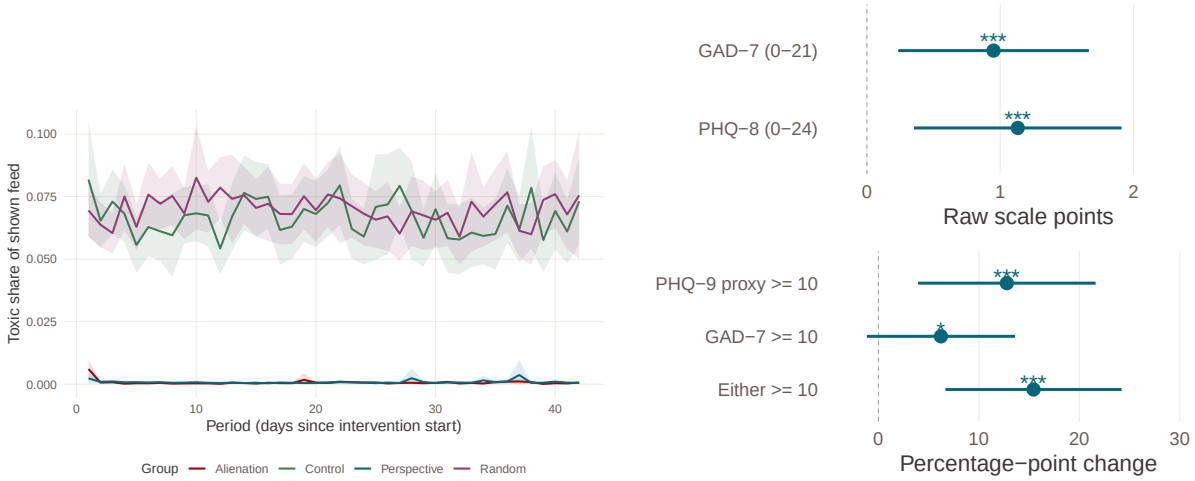
(A) Clinical scales (B) Moderate-symptom threshold

Figure 3: Main Effect

Notes: The figure reports ITT estimates from the specification in Equation (1) on anxiety (GAD-7) and depression (PHQ-9 proxy) scales, as well as the crossings of participants into the scores that are classified as moderate depression or anxiety. Bars denote 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Item-level decomposition. Item-level results are presented in the Online Appendix. For the GAD-7, coefficients are positive across all seven items. The largest increases are on feeling afraid as if something awful might happen (GAD7), feeling nervous or anxious (GAD1), trouble relaxing (GAD4), and becoming easily annoyed or irritable (GAD6). The PHQ-8 coefficients are largest for trouble falling or staying asleep (PHQ3), feeling down, depressed, or hopeless (PHQ2), and feeling bad about oneself or like a failure (PHQ6). Both scales show modest increases across several symptoms, and no single item drives the result.

Welfare costs. Our calibration gives a welfare cost of approximately $158 per treated participant over six weeks. We use the PHQ-8 and GAD-7 ITT estimates and each participant's pre-treatment scores to compute the implied health loss. We then convert it into quality-adjusted life-years (QALYs), which weight time by health-based quality of life. We value a QALY at each participant's reported household income and average the resulting dollar costs. The Online Appendix describes the mapping from symptom scores to utility and the calculation.



(A) Shown toxic share (B) Exposure-scaled coefficients

Figure 4: Exposure-Scaled Effect

Notes: Panel (a) shows the share of shown content classified as toxic in each arm over the 42-day intervention. Panel (b) reports the mental-health ITT estimates rescaled to a 10 pp reduction in this share. Positive values denote an increase in mental health symptoms as exposure to toxicity falls. The upper part reports changes in GAD-7 and PHQ-8 scores in raw points. The lower part reports changes in the share of respondents classified at or above the cutoff for moderate symptoms. Bars show 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

4.2 Treatment Intensity and Exposure-Scaled Effects

Among treated participants, the extension hid an average of 54 content items per active day, including approximately 33 posts and 21 comments. Over the 42-day intervention, the toxic share of shown content averages 7.75% in control and 0.05% in treatment. The estimated first stage is a 7.81 pp reduction.[22] The arm-level pattern in Figure 4a makes the same point: pure control and random hiding yield shares of toxic content shown equal to 7.63% and 7.86%, while the Perspective and Perspective+Alienation treatment arms give 0.07% and 0.03%. We conclude that the intervention led to a meaningful reduction in exposure to toxic content.

To further evaluate the direct impact of exposure to toxic content, we scale the reduced form by the first stage in Figure 4b. A 10 pp reduction in shown toxic share corresponds to an increase of 0.19σ in the index of mental health symptoms. On the raw clinical scales, the same reduction corresponds to 1.0 GAD-7 points and 1.1 PHQ-8 points.

4.3 Benchmarking Magnitude

To place the magnitude of our headline ITT in context, we compare it with intended benefits to mental health reported in four experimental studies: a mindfulness app, an AI-powered cognitive behavioral therapy app, behavioral-activation psychotherapy, and a one-time psychedelic treatment. Table 2 presents the summary of each intervention. Our effect is between 33% and 65% of comparable interventions, in the opposite direction.

4.4 Heterogeneous Treatment Effects

Our results are suggestive of toxicity filtering worsening mental health more among respondents who reported more symptoms before treatment. Figure 5 illustrates this pattern using an exploratory split into quartiles. The ITT estimates rise across the four quartiles, from 0.05σ among respondents with the fewest pre-treatment symptoms to 0.22σ among those with the most. Having said that, the difference between the highest and lowest quartiles is not statistically significant, so we interpret this pattern with caution. The preregistered median split gives the same ordering. The ITT on the index of mental health symptoms is 0.21σ for respondents above the median, compared with 0.09σ for those below it. The difference is 0.12σ, but it is also imprecisely estimated ($p = 0.254$).

[22] On Reddit, hiding a toxic comment also hides its replies, but the logs still count those replies as shown, introducing a small measurement error in the toxic shares. Because the first stage is large, this does not meaningfully affect our estimates. Moreover, if these replies are more toxic than the content that remains visible, the logs overstate the toxic share shown to treated participants, understating the reduction in their exposure.

Table 2: Benchmarking the ITT against mental-health interventions

Study	Intervention	Improvement (σ)	Horizon	Our ITT as share of benchmark
This paper	Toxicity filtering on social media	−0.15	6 weeks	–
Shreekumar and Vautrey (2024)	Headspace mindfulness app	0.46	4 weeks	33%
Angelucci et al. (2026)	AI-powered CBT app	0.29	6 months	52%
Bhat et al. (2022)	Behavioral-activation psychotherapy	0.23	2 brief trials, 4–5 years later	65%
Francois et al. (2025)	One-time ayahuasca treatment	≈0.4	6 months	38%

Notes: Shreekumar and Vautrey (2024): participants randomized to receive access to the Headspace app; the outcome is an index that aggregates validated measures of anxiety, depression, and stress, which falls by 0.38σ after two weeks and by 0.46σ after four weeks. Angelucci et al. (2026): 1,964 Mexican women with mild to severe psychological distress randomized to free access to an AI-powered, CBT-based mental health app for up to six months; the primary scales are PHQ-8 and GAD-7 – the same instruments as our study – and the 0.29σ improvement in a composite mental health index at the six-month endpoint is a magnitude the authors note is comparable to psychotherapy and pharmacotherapy. Bhat et al. (2022): long-run follow-up of two brief, inexpensive behavioral-activation psychotherapy trials in India; the table reports the PHQ-9 depression reduction for the most effective arm (the Healthy Activity Program), which also cut moderate-symptom prevalence by 13 percentage points; pooled across both trials the reduction is 0.15σ. Francois et al. (2025): the largest randomized controlled trial of psychedelics to date, partnering with an ayahuasca center in Brazil to randomize 429 first-time users to a one-time ayahuasca treatment or a ritualized placebo; relative to placebo, ayahuasca raises happiness and lowers psychological distress six months later, with the gains concentrated among participants who were distressed at baseline.

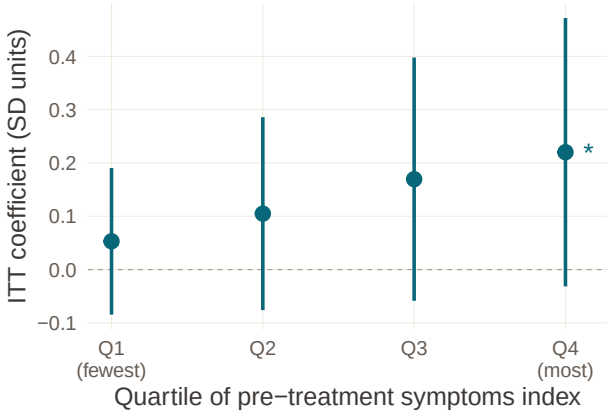


Figure 5: Heterogeneity by pre-treatment mental-health

Notes: This figure reports estimates of the effect of the intervention on the standardized mental-health-symptoms index, separately by quartile of pre-treatment mental health. Points denote quartile-specific intention-to-treat (ITT) coefficients, where quartiles are defined using pre-treatment symptoms. Bars denote 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

In addition, we find little evidence that the ITT differs by sex, age, education, ethnicity, income, or use intensity. The Online Appendix reports the full median-split results, quartile estimates based on the anxiety and depression scales, and the remaining subgroup comparisons.

5 Mechanisms Evidence

The adverse mental-health ITT is consistent with the view of social media as a forum in which users maintain social ties and participate in shared discussions (Enikolopov et al., 2024; Donati and Song, 2026; Zhuravskaya et al., 2020). Even a hostile post can let users hear from peers, see that others share their concerns, or find others defending causes they care about. Filtering these exchanges may therefore remove some of the social benefits of platform use and leave users feeling less connected to others, despite preserving access to the platforms.

The secondary survey outcomes and ITT heterogeneity support this interpretation. Filtering increases loneliness, and the adverse mental-health ITT is larger on Facebook and X (more oriented toward social connections) than on Reddit (more oriented toward information). The adverse ITT is also larger among users with more peer content in their feeds and those with stronger political attachments. Peer share and pre-treatment hostility toward the opposing party remain associated with a larger adverse ITT when included in the same specification. These patterns suggest that filtering is more costly when the hidden content comes from, or speaks for, "your people."

5.1 Measures

5.1.1 Secondary Outcomes

We organize the secondary outcomes into four groups. Two groups are predicted ex ante to be *negative mediators* – channels through which toxicity filtering could worsen mental health: moral self-view and social experiences. The third group – social beliefs – is predicted ex ante to be a *positive mediator* – a channel through which filtering could improve mental health. Lastly, we report behavioral adaptation based on willingness to follow toxic accounts and substitution to other social media.

Moral self-view. Seeing others behave badly on social media may make users feel better about their own conduct (Wills, 1981). If users value this comparison, filtering toxic content could weaken their sense of moral superiority and worsen mental health (Tappin and McKay, 2017; Taylor and Brown, 1988). We examine this channel using two measures. The *absolute moral superiority index* combines five items from the Marlowe–Crowne Social Desirability Scale about respondents' own attitudes and behavior. The *relative moral superiority measure* asks whether respondents see themselves as having more or fewer positive qualities than the average person.

Social experiences. Even hostile exchanges on social media may help users feel connected to others and keep up with discussions they care about. Filtering such content could leave users feeling isolated or worried about missing discussions they care about, potentially worsening mental health. We examine this channel using measures of loneliness, fear of missing out (FOMO).[23] We elicit loneliness using two items from the UCLA Loneliness Scale, asking how often respondents lack companionship or feel left out (Hughes et al., 2004). A separate question asks whether social media makes respondents feel they are missing important information, experiences, or discussions (Bursztyn et al., 2025). Finally, we ask how often respondents feel angry, to capture the potential of consuming toxic content to release anger.

Social beliefs. Filtering toxic content may make discrimination seem less common, which could improve mental health (Pascoe and Richman, 2009). We measure perceived discrimination by describing six scenarios involving gender, race, sexual orientation, and age, and eliciting beliefs about how often each occurs.[24] We also report respondent-relevant discrimination, using only the scenarios that apply to each respondent's demographic characteristics. To measure perceived altruism, we elicit beliefs about the fraction of a bonus payment that participants in an unrelated study donated to Save the Children USA.

Behavioral adaptation. We also examine whether users respond to filtering by seeking out toxic content or spending more time on other (untreated) social media platforms. In the post-treatment and follow-up surveys, we elicit the desire to follow politically toxic content with a list experiment. We cross-randomize which accounts respondents see: three neutral X accounts, or those same accounts plus one posting politically toxic content, and respondents then report the number of accounts they would like to follow. This technique allows us to estimate interest in the toxic account while reducing social-desirability bias (Bursztyn et al., 2019). We also use data from Beknazar-Yuzbashev et al. (2025) to assess whether increased use of platforms outside the intervention could account for a deterioration in mental health.

5.1.2 Pre-Treatment Political Covariates

We examine heterogeneity in the mental-health ITT by political characteristics measured in the pre-treatment survey: party affiliation and political-hostility indices.[25] Our central measure of political hostility is the moral disengagement index (MDI), which averages agreement

[23] Social-experience outcomes were added to the survey after the post-treatment survey of Wave 1.

[24] The wording of the scenarios is provided in the Online Appendix

[25] The pre-treatment survey carries political outcomes for a separately preregistered political-attitudes study (see Section 2.3). In the present paper these measures enter exclusively as pre-treatment covariates to study heterogeneity of treatment effects on mental health.

with three statements about the respondent's political out-party – that it is "a serious threat to the United States and its people," that it is "not just worse for politics" but "downright evil," and that many of its members "lack the traits to be considered fully human" – each rated on a seven-point agreement scale adapted from Kalmoe and Mason (2022). High values capture a morally charged relation to the political out-party: viewing it as threatening, evil, or less than fully human. Despite the somewhat extreme nature of the statements, the shares of respondents agreeing with them are considerable – 50% considers the out-party a threat to the U.S., 30% thinks it is downright evil, and 18% questions humanity of the out-party members (the full distribution is in the Online Appendix). We use other indices as comparison dimensions in Section 5.4, which include the dehumanization index, the political polarization index, the support-for-political-violence index, and the spiral-of-silence index (see the Online Appendix for their definitions).

5.1.3 Platform Use and Feed Composition

Proxies for the pre-treatment feed environment. To study how the mental-health ITT varies with users' social media environments, it is helpful to proxy four pre-treatment features of a user's social media environment: platform use, toxicity of supplied content, author composition of the feed, and partisan orientation of toxic content. We do not have a baseline period (Section 2.5), hence we construct proxies from the extension's post-randomization activity logs, recognizing that treatment may itself affect these measures.

We limit the risk of endogeneity in multiple ways. For content-based measures, we use *supplied*, pre-hiding feed posts. We drop comments because their delivery depends on which items the user chooses to open and is therefore not determined solely by the platform. Moreover, for measures that are likely to respond quickly to the treatment, we rely only on the first qualifying days after treatment begins (the early window).[26] Details are provided in the Online Appendix, alongside balance across treatment arms and evidence that the results are robust to alternative construction choices.

Platform mix and toxicity of supplied content. The platform mix is the share of a user's logged platform time spent on Facebook, Reddit, and X, averaged over the early window. We define supplied-content toxicity as the daily share of main-feed posts served to the user that score above the 0.33 Perspective threshold, averaged over the early window.

[26] We measure platform use and supplied-content toxicity over the first five qualifying days within two weeks of treatment start (requiring at least three), before users adjust to the unannounced filter. Peer share and the partisan orientation of toxic content are less likely to respond quickly to treatment and require more observations, so we compute both over the full logging period.

Peer share. The peer share is the share of supplied post impressions from *horizontal* accounts – personal accounts classified as peers – as opposed to *vertical* accounts, such as pages, public figures, and other broadcast sources.[27] We use posts supplied before filtering and construct the measure for Facebook and X. We exclude Reddit, as the platform's feeds are organized around communities rather than individual authors. We weight each day's platform-specific shares by the user's early platform mix, and then we average these daily shares over the logging window.

Toxic political content. We also examine whether toxic political content in users' feeds favors their own party or the opposing party. We use a large language model to identify political posts among the toxic posts supplied on Facebook, Reddit, and X, and classify which party they favor. An attack on one party is coded as favoring the other. Mixed or unclear cases are labeled ambiguous. We combine these labels with users' pre-treatment party affiliation to examine how closely this content aligns with their own political side.

5.2 Does Filtering Leave Users Feeling Less Connected?

Figure 6a presents preregistered secondary outcomes. Toxicity filtering significantly raises the index of loneliness, by 0.34σ ($p = 0.003$),[28] while anger, FOMO, social desirability, perceived discrimination, and altruism move little.[29] Details and Benjamini–Hochberg corrections across the secondary-outcome family are reported in the Online Appendix. Crucially, loneliness, our main mechanism, survives the correction.

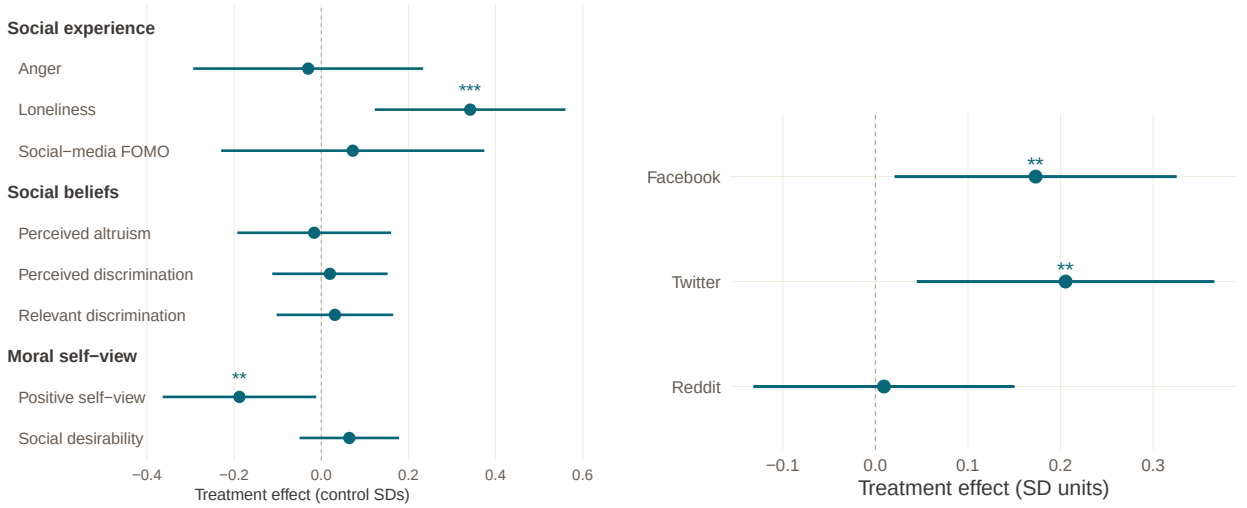
The increase in loneliness supports the forum view of social media. Following a conversation can make users feel connected to others who share their interests or concerns, even when the exchange is hostile. Filtering such exchanges may therefore weaken the sense of participation that users derive from the platform.

The question then becomes, which features of users' social environments are associated

[27] Accounts are classified using their reach across study participants, typical engagement, and verification status. The Online Appendix provides the classification rules.

[28] This is the six-week post-treatment estimate, identified from the preregistered within-person comparison between the pre-treatment and six-week surveys (342 observations). These items were introduced by the October 2024 preregistration amendment, so only the second recruitment wave answered them at the six-week survey; the first wave answered them only at the twelve-week follow-up and therefore contributes a cross-section rather than a within-person comparison. Pooling that cross-section with the six-week estimate by inverse-variance weighting gives a smaller but still significant 0.21σ ($p = 0.005$) effect, statistically indistinguishable from the six-week estimate (equality test $p = 0.121$).

[29] Treated respondents also assess their own positive qualities less favorably relative to the average person. This is consistent with the social-comparison channel discussed in the introduction: seeing others behave badly may make users judge themselves more favorably. The ITT is no longer statistically significant after Benjamini–Hochberg correction across the secondary outcomes, so we treat this evidence as suggestive.



(A) Secondary outcomes (B) Platform heterogeneity

Figure 6: Secondary outcomes and platform heterogeneity

Notes: This figure reports estimates of the effect of the intervention on secondary outcomes (Panel (a)) and the implied per-platform effects on mental health (Panel (b)). Panel (a) reports intention-to-treat (ITT) coefficients on the secondary survey outcomes, standardized in control-group standard-deviation units. Panel (b) reports implied per-platform ITT coefficients on the standardized mental-health-symptoms index, obtained from the early-platform-mix specification, with Facebook as the omitted platform. Bars denote 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

with greater costs from filtering?

5.3 Does the Cost Vary with Peer Content?

The implied mental-health ITTs are positive for Facebook and X and close to zero for Reddit (Figure 6b). On Facebook and X, posts and comments are more often part of ongoing conversations involving people users know or follow. Reddit is organized more around topics and information search. This pattern is consistent with filtering being more costly when content helps users maintain social connections.

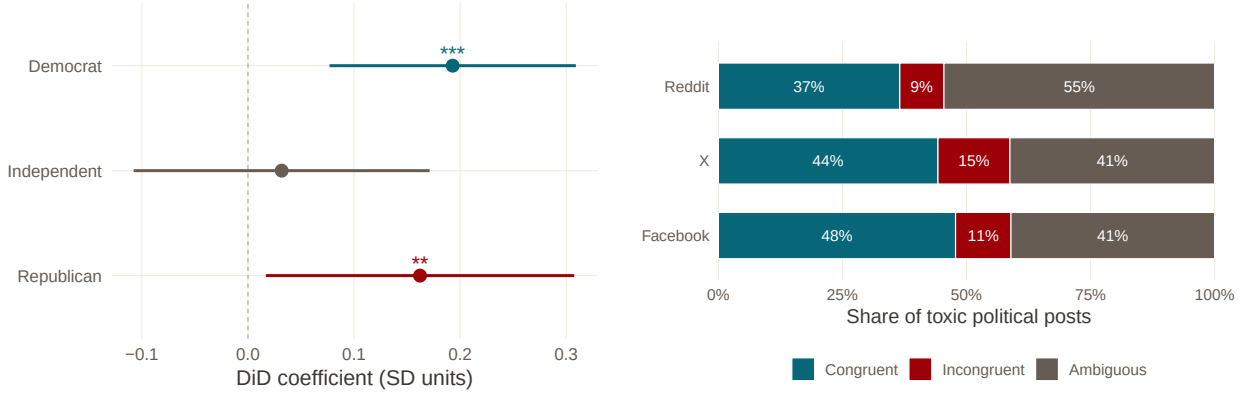
Since platforms differ along several dimensions, we next examine whether the adverse ITT varies with the amount of peer content conditional on platform mix. We measure peer share as the share of supplied feed impressions from personal accounts classified as peers.[30] Conditional on early platform shares, a one SD increase in peer share is associated with a 0.11σ larger adverse effect on mental health ($p = 0.013$; Table 3, column 1). Even hostile posts from these accounts may help users keep up with exchanges in their social environment. The larger adverse ITT among users with more peer content is therefore consistent with

[30] Accounts are classified using their reach across study participants, typical engagement, and verification status (Section 5.1.3). This classification proxies for horizontal ties without directly measuring personal relationships between users and authors.

filtering removing socially valued exchanges.

5.4 Does Political Attachment Matter?

Content may also connect users to others through shared political concerns, including when its author is someone they do not know personally.[31] For users who identify with a political party, even hostile exchanges may provide a sense that others share their concerns or support their side. The estimates by party affiliation are consistent with this possibility: the mental-health ITT coefficients are positive for Democrats and Republicans and close to zero for independents (Figure 7a).



(A) Party-specific ITT coefficients (B) Toxic political content by platform

Figure 7: Political attachment and toxic political content

Notes: Panel (a) reports ITT estimates for the standardized mental-health-symptoms index, separately by party affiliation. Panel (b) shows post-weighted shares by platform for toxic political posts recorded during treatment days 1–42. It includes respondents who identify with or lean toward the Democratic or Republican party. An attack on one party is coded as favoring the other. Congruent posts favor the respondent's party, and incongruent posts favor the opposing party. We label all other cases ambiguous, including mixed, non-U.S., and unclear partisan content. Bars in panel (a) show 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

The partisan orientation of toxic content helps explain why its removal could impose these costs. Among toxic political posts with a discernible affiliation, most favor the user's own party (Figure 7b). Such posts may express hostility toward political opponents while also showing the user that others share their concerns. Filtering them may therefore reduce opportunities to see those concerns voiced and supported.

Users who view the opposing party as threatening or morally objectionable may particularly value posts showing that others share their concerns. We examine this possibility using the pre-treatment moral disengagement index (MDI), which measures agreement that

[31] We focus on politics because partisan identity shapes behavior beyond political settings (Iyengar et al., 2012) and influences online tie formation (Mosleh et al., 2024).

the opposing party is threatening, evil, or lacking humanity.[32] Higher MDI is associated with a larger adverse mental-health ITT (Table 3, column 2). This pattern is consistent with filtering being more costly for users who place greater value on expressions of shared political concern.

5.5 How Do Social and Political Proximity Fit Together?

The preceding results associate a larger adverse mental-health ITT with both a higher share of peer content and greater hostility toward the opposing party. These patterns could overlap: users with stronger political hostility may also encounter more peer content. We therefore include peer share and pre-treatment MDI jointly to examine whether each remains associated with ITT heterogeneity after accounting for the other. Columns 3–5 of Table 3 report these joint specifications, using the same sample as the separate specifications in columns 1–2.

Table 3: Peer-share and MDI heterogeneity combined

	(1)	(2)	(3)	(4)	(5)
Treated × peer share	0.110**		0.140***	0.141***	0.141***
	(0.044)		(0.045)	(0.046)	(0.046)
Treated × MDI		0.116***	0.129***	0.128***	0.128***
		(0.042)	(0.042)	(0.042)	(0.042)
Treated × peer share × MDI				0.023	0.023
				(0.042)	(0.042)
Observations	862	862	862	862	862
Baseline symptoms controls		✓	✓	✓	✓
Platform share controls	✓		✓	✓	✓
Toxicity-dose control					✓

Notes: This table reports estimates from six-week post-treatment difference-in-differences regressions for the standardized index of mental health symptoms. All columns use the common sample of 431 participants for which all the variables are available and include ID and cohort-by-time fixed effects. The main ITT specification without heterogeneity interactions gives $0.18\,\sigma$ on this sample (standard error 0.06). Peer share and the pre-treatment moral disengagement index (MDI) are standardized. Standard errors clustered by respondent are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

In the joint specification, a one-standard-deviation increase in peer share is associated with a $0.14\,\sigma$-standard-deviation larger adverse ITT, while the corresponding coefficient for MDI is $0.13\,\sigma$; both are statistically significant (column 3). Each association therefore remains after accounting for the other dimension, platform mix, and pre-treatment symptoms. The coefficients are essentially unchanged when the specification additionally allows the ITT

[32] We also examine other political measures: political polarization, support for political violence, dehumanization, and spiral of silence – their coefficients are smaller and not statistically distinguishable from zero (reported in the Online Appendix).

to vary with the early share of supplied posts classified as toxic (column 5), suggesting that the two patterns are not explained by this measure of feed toxicity. We find no statistically detectable interaction between peer share and MDI, although that estimate is imprecise (column 4).

Together, the joint estimates suggest that both the composition of users' feeds and their political concerns help account for differences in the mental-health costs of filtering.

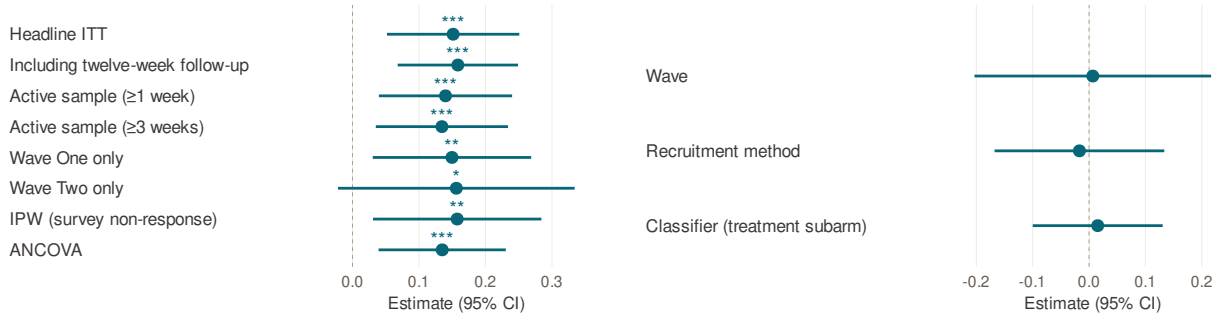
6 Robustness and Alternative Interpretations

We conduct a battery of robustness checks to strengthen the internal validity of our estimates. Section 6.1 considers whether the treatment effects on mental health are robust to alternative study horizons, sample selection, weighting, and implementation. It also provides evidence that our results are unlikely to be explained by differential attrition, pre-treatment imbalance among survey respondents, or multiple hypothesis testing. Section 6.2 discusses the validity of our interpretation of the results and addresses alternative explanations.

6.1 Robustness of the Estimates

Alternative specifications. Figure 8 reports the ITT effect on the index of mental health symptoms under several alternative specifications, which include adding the twelve-week follow-up survey to the estimation sample, estimating the coefficient among respondents who remained active extension users for longer windows, and allowing the coefficient to differ by recruitment wave, recruitment source, and toxicity classification method.

Panel A of Figure 8 shows that the treatment effect is close to the headline ITT when we add the twelve-week follow-up, so the headline result is not specific to the preregistered six-week endpoint. Moreover, the effect is robust to restricting attention to respondents who remained active extension users for at least one or three weeks. This implies that the result is not driven by users who installed the extension and immediately stopped using it. The interaction coefficients reported in Panel B of Figure 8 also indicate that the effect does not depend on the recruitment wave (during an election campaign or outside of it), recruitment method (CloudResearch vs. social media ads), or toxicity classification (Perspective+Alienation vs. Perspective). All four of the coefficients are insignificant and close to zero. With respect to recruitment waves, the red coefficients in Panel A of Figure 8 show that the treatment effect is of the same sign and significant when separately estimated for each wave (0.15σ and 0.16σ). We conclude that our headline result replicates in two distinct data collections conducted at different times and in different political environments.



(A) Treated (B) Treated × subgroup

Figure 8: Robustness panel

Notes: This figure reports difference-in-differences estimates of the effect of the intervention on the standardized index of mental health symptoms, where each row is a separate estimate and higher values indicate worse mental health. Panel (a) reports the treated coefficient under alternative specifications: the headline ITT; adding the twelve-week follow-up survey to the estimation sample; restricting to respondents who remained active extension users for at least one or three weeks; estimating the ITT separately for each recruitment wave; reweighting respondents to the post-treatment survey by the inverse of their predicted completion probability; and an ANCOVA specification that regresses the post-treatment index on treatment assignment and the pre-treatment index with start-date cohort fixed effects and heteroskedasticity-robust standard errors. For the rows pertaining to the active samples, the ITT is the sum of the treated coefficient and its interaction with the activity indicator, and its standard error is computed from the covariance matrix clustered at the respondent level. Panel (b) reports the interaction coefficient (treated × subgroup) based on specifications where the treatment effect can differ by recruitment wave, recruitment method (CloudResearch vs. social media ads), and toxicity classification (Perspective+Alienation vs. Perspective). Bars denote 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Attrition. We consider two margins of potential differential attrition. The first is survey non-response: the post-treatment survey outcomes are observed only for respondents who complete that survey. The second is participants stopping extension use. Table 4 studies both margins among participants who completed the pre-treatment survey and recorded any extension activity after assignment ($N = 1270$). The left block uses post-treatment survey completion as the dependent variable. The right block uses an indicator for remaining active on the extension through post-treatment day 42, which approximately coincides with the time of the post-treatment survey.

Table 4 indicates that treatment assignment does not meaningfully predict post-treatment survey completion. In particular, without controls, the difference in survey completion between treatment and control participants is less than 1 pp and insignificant (column 1). The treatment coefficient remains insignificant as the specifications progressively add pre-treatment symptoms, allow the treatment coefficient to vary with symptom severity, and then add demographic controls and treatment-by-demographic interactions. Extension drop-out rates also do not vary by treatment across the five specifications. The joint Wald tests in the interaction columns likewise do not reject the null hypothesis of no differential attrition for either outcome.

In the Online Appendix, we provide further evidence to address concerns about differen-

Table 4: Differential attrition

	Completed post-treatment survey					Still using extension at day 42				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Constant	0.517***	0.551***	0.521***	0.432***	0.399***	0.785***	0.797***	0.814***	0.624***	0.540***
	(0.023)	(0.027)	(0.032)	(0.071)	(0.113)	(0.019)	(0.021)	(0.025)	(0.069)	(0.107)
Treatment	0.009	0.008	0.058	0.044	0.106	−0.026	−0.026	−0.054	−0.043	0.088
	(0.029)	(0.029)	(0.041)	(0.036)	(0.140)	(0.024)	(0.024)	(0.033)	(0.033)	(0.135)
N	1,270	1,270	1,270	1,262	1,262	1,270	1,270	1,270	1,262	1,262
Baseline symptoms		✓	✓	✓	✓		✓	✓	✓	✓
Baseline symptoms × Treated			✓	✓	✓			✓	✓	✓
Demographics				✓	✓				✓	✓
Demographics × Treated					✓					✓
Joint F-stat (Treatment + Treatment×X = 0)			1.521	1.233	0.926			1.337	0.863	0.665
Joint p-value			0.219	0.292	0.508			0.263	0.422	0.757

Notes: This table reports OLS estimates (linear probability model) of differential attrition among participants who completed the pre-treatment survey and recorded any extension activity after treatment assignment. The dependent variable is post-treatment survey completion in the left block and an indicator for remaining an extension user through post-treatment day 42 in the right block. For each outcome, we add controls cumulatively: treatment only; an indicator for above-median pre-treatment symptoms; its interaction with treatment; demographic controls; and their interactions with treatment. Checkmarks indicate the included control variables. In columns (1)–(2) and (6)–(7), the treatment coefficient corresponds to the treated–control difference in the outcome. In the remaining columns, it is the difference among participants with below-median pre-treatment symptoms, evaluated at zero values of the demographic controls where these are included. Columns (4)–(5) and (9)–(10) drop participants with missing demographics. The joint Wald rows test the null hypothesis that the treatment coefficient and all treatment-by-covariate interactions are jointly zero. Heteroskedasticity-robust standard errors are parenthesized. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

tial attrition. First, we show that the trajectory of extension use drop-out over time does not meaningfully vary across experimental conditions. Second, we detect no differential attrition in post-treatment survey completion when we compare the two treatment arms (Perspective + Alienation vs. Perspective) and the two controls arms (random vs. pure). Third, we find no differential attrition in completion of the follow-up survey (at twelve weeks). Taken together with Table 4, we conclude that differential attrition is unlikely to explain our results.

Composition of survey respondents. The lack of differential attrition rates does not rule out that participants who completed the post-treatment survey differ in type from those who did not. We consider this risk by reweighting respondents to the post-treatment survey by the inverse of their predicted completion probability (see Panel A in Figure 8). This regression yields a significant treatment effect of similar magnitude (0.16σ) to the headline ITT (0.15σ). The effect is also robust to fitting the IPW completion model separately for treatment and control arms (0.13σ, s.e. 0.06). We conclude that selection into completing the post-treatment survey on observed pre-treatment covariates, whether common to both arms or differential by arm, is limited.

A separate concern is that, despite balanced randomization (Section 2.4), treated respondents to the post-treatment survey had fewer mental health symptoms pre-treatment than control respondents (-0.11σ, $p = 0.148$), so that mean reversion could raise their symptoms over the study period. We address this by estimating an ANCOVA specification, which, unlike the difference-in-differences strategy, would not overestimate the ITT in the presence of

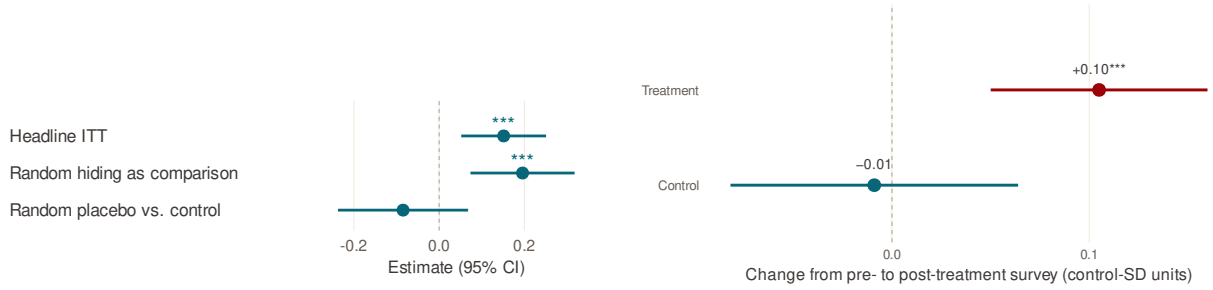
mean reversion (McKenzie, 2012; Ding and Li, 2019). The treatment effect remains significant and of similar magnitude under ANCOVA (0.13σ, s.e. 0.05). We also estimate our main specification with cohort-time fixed effects that vary by pre-treatment symptom quartile, which compares treated and control participants with similar initial symptoms, and find a significant effect (0.14σ, $p = 0.009$), further addressing the issue.

Multiple hypothesis testing. We apply the Benjamini and Hochberg (1995) correction across the three preregistered primary outcomes (the PHQ-8, the GAD-7, and their composite). Under the registered specification, all three effects remain significant after the correction ($q = 0.010$, $q = 0.007$, and $q = 0.007$, respectively). Because our experimental infrastructure also served a separately preregistered political attitudes study, we also consider families that include its outcomes. Adding its 4 primary outcomes, the largest possible q-value for the index of mental health symptoms in the resulting 5-outcome family is $q = 0.015$ (unadjusted $p = 0.003$). In a broader family of 16 hypotheses that also includes political attitude outcomes from the common survey instrument and our nine secondary outcomes, both the index of mental health symptoms and loneliness have $q = 0.024$. Even in the worst case, in which every added outcome has a larger p-value, both would remain significant at the 5% level after adding 17 further outcomes to this broad family ($q = 0.049$) and would cross the threshold only after adding 18 ($q = 0.051$). We conclude that our results are not driven by multiple hypothesis testing.

6.2 Validity of the Interpretation

User experience and feed composition. The intervention reduces exposure to toxic content, but there is a natural concern that it might affect mental health through other channels, hiding content per se and changing the feed composition. The random hiding control addresses the former concern directly by hiding a matched fraction of content without targeting toxicity. Relative to pure control, the random hiding coefficient on the index of mental health symptoms is statistically insignificant, and of the opposite sign to the treatment effect (Panel A of Figure 9). When the random hiding arm is used as the comparison group, the ITT remains positive (0.20σ) and of similar magnitude to the headline ITT.

We also report that the intervention does not change the topical composition of users' feeds. We classify all posts shown to participants in the analysis sample during the six-week treatment period into 25 topic categories adapted from the IAB Content Taxonomy. We find no significant treatment effect on 24 of the 25 categories, and the effect on the remaining category (sports) does not survive the Benjamini and Hochberg (1995) correction. A joint test does not reject the null hypothesis that all differences are zero ($p = 0.696$). In the Online



(A) Changes to user experience (B) Within-arm changes

Figure 9: Additional Robustness

Notes: This figure reports estimates for the standardized index of mental health symptoms, where higher values indicate worse mental health. Panel (a) reports difference-in-differences estimates of the effect of the intervention, where each row is a separate estimate, for three specifications: the headline ITT; using the random hiding control as the comparison group; comparing the random hiding control with the pure control. Bars denote 95% confidence intervals based on standard errors clustered at the respondent level. Panel (b) reports the mean change in the index between the pre-treatment and post-treatment surveys, separately for control and treatment participants in the analysis sample. Bars denote 95% confidence intervals for the mean change within each group. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix, we report the coefficients for all categories and show that they are an order of magnitude smaller than the corresponding coefficient for toxic content.

We find no evidence that the results are driven by the respondents in the treatment group experiencing a noticeable change in their user experience or realizing the exact purpose and functionality of the extension. We explicitly asked the respondents whether they had experienced any change in their social media experience during the weeks of the experiment, and we find no evidence that treated respondents reported more change: on a five-point scale, the treated–control gap is small and insignificant (-0.03, s.e. 0.10, $p = 0.729$).[33] This finding also mitigates concerns that the effect might be due to experimenter demand effects or perceived loss of agency over one's social media feed.

In addition to showing that the intervention is unlikely to affect mental health through channels other than toxicity exposure, we check whether the effect reflects an improvement in the control group rather than a deterioration in the treatment group, for instance due to an unobserved shock. Panel B of Figure 9 shows that mental health symptoms in the control group did not change between the pre-treatment and post-treatment surveys (-0.01σ, $p = 0.810$), whereas symptoms in the treatment group rose by 0.10σ ($p < 0.001$).

Transitory adjustments. We also address the concern that users may initially react to the sudden removal of familiar content and then revert as they adapt. Figure 10 shows

[33]This survey question was introduced partway through data collection and has no pre-treatment counterpart, so we pool all post-treatment responses – both the six-week and twelve-week surveys and both recruitment waves – to maximize power, comparing treated and control respondents within cells defined by start-date cohorts and survey and with standard errors clustered at the respondent level.

the ITT coefficients separately at the post-treatment and follow-up surveys. Using the unbalanced panel (users who took the pre-treatment survey and at least one of the later surveys), the symptom index rises by 0.16σ at six weeks and remains elevated by 0.11σ at twelve weeks. Since this analysis may be affected by changes in the composition of respondents across the post-treatment and follow-up surveys, we also plot the effects for the balanced panel of respondents who completed all three surveys. Although the balanced panel is smaller and its estimates less precise, the adverse effect is detectable at six weeks (0.16σ) and most of it persists at twelve weeks (0.10σ). Taken together, the two panels are difficult to reconcile with a purely transitory adjustment response.

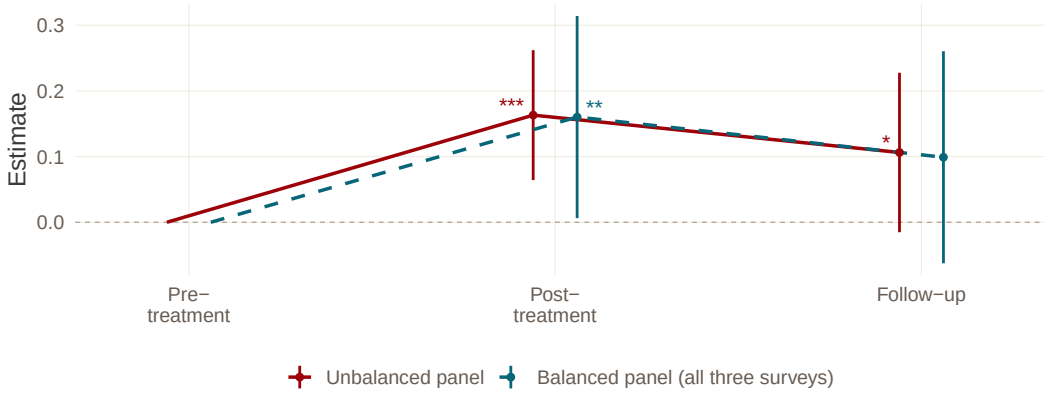


FIGURE 10: DYNAMICS OF THE EFFECT ON MENTAL HEALTH

Notes: This figure reports difference-in-differences estimates of the effect of the intervention on the standardized index of mental health symptoms at the six-week post-treatment survey and the twelve-week follow-up survey, with the pre-treatment period normalized to zero. Higher values indicate worse mental health. We report estimates for two samples. The first one uses the unbalanced panel, which includes all participants who completed the pre-treatment survey and at least one of the later surveys. The second one uses the balanced panel, which restricts the analysis sample to participants who completed all three surveys (pre-treatment, post-treatment, and follow-up). Both samples only include participants for whom we recorded at least some browser extension activity. Bars denote 95% confidence intervals based on standard errors clustered at the respondent level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

A related concern is that our results may be explained by users experiencing a withdrawal response typical of addictive goods. Given that our headline estimates are measured at six weeks and persist at twelve weeks, it is difficult to reconcile our results with the withdrawal durations of substances such as caffeine or nicotine. In the Online Appendix, we provide evidence that for addictive substances such as nicotine, alcohol, cannabis, and caffeine, the peak of withdrawal symptoms occurs within days and the acute symptoms resolve within four weeks. Moreover, even for nicotine, which has a longer withdrawal duration than the other three substances, mental health among quitters is better than among continuing smokers by seven weeks after cessation (Taylor et al., 2014). Toxic content plausibly creates a weaker dependence than nicotine, so if anything the comparison understates how early we

should expect to see a sign reversal.[34] We regard this as evidence against the withdrawal interpretation of our results.

Classifier validation. We address the concern that the intervention hides content that is not genuinely toxic due to classification error. Even though Perspective API is a commercial-grade classifier that has been widely applied in research and business settings, we validate it against two external classifiers: (1) a large language model classifying uncivil content using a coding rubric adapted from Coe et al. (2014), and (2) OpenAI's moderation model classifying harassing and hateful content (OpenAI, 2026). Figure 11 presents the results. Although neither classifier is designed to capture all toxicity, which Perspective defines through crowd annotators' judgments, 70% of posts above the Perspective hiding threshold are flagged as either uncivil or harassing or hateful. The right panel of Figure 11 supports including content with toxicity scores between 0.33 and 0.70 in the hiding rule. 78% of posts in the 0.50–0.70 range and 55% of posts in the 0.33–0.50 range are classified as uncivil. Even for the more extreme category of harassing or hateful content, 45% of posts in the 0.50–0.70 range and 21% of posts in the 0.33–0.50 range are flagged. This is consistent with the analysis of examples of toxic content in Beknazar-Yuzbashev et al. (2025), which indicates that much harmful toxic content has scores between 0.33 and 0.70.

Substitution. Another concern relates to substitution outside the treated desktop feeds. If users replace hidden toxic content with equally or more toxic content elsewhere, the intervention may not reduce their overall toxicity exposure and could, in the extreme, raise it. We examine this concern in three ways. First, we show that treatment does not shift reported use from desktop to unfiltered mobile devices. Second, while prior evidence using an analogous extension shows some substitution toward other social media sites (Beknazar-Yuzbashev et al., 2025), that substitution does not go toward sites classified as more toxic. Third, in a list experiment in the post-treatment survey, treated respondents do not report a statistically significant increase in willingness to follow a politically toxic account. The full substitution evidence is reported in the Online Appendix.

Algorithmic response. Lastly, we do not find a statistically significant difference in the share of toxicity that platforms supply (before filtering) between treatment and control, suggesting a lack of algorithm adaptation. The companion study on engagement using a

[34] Substances inducing milder dependence have shorter and less severe withdrawals. Withdrawal from caffeine, which induces weaker dependence, resolves within two to nine days (Juliano and Griffiths, 2004), compared with two to four weeks for nicotine (Hughes, 2007). Within a substance, withdrawal is more severe and can last longer with heavier use (Baker et al., 2012; Connor et al., 2022).

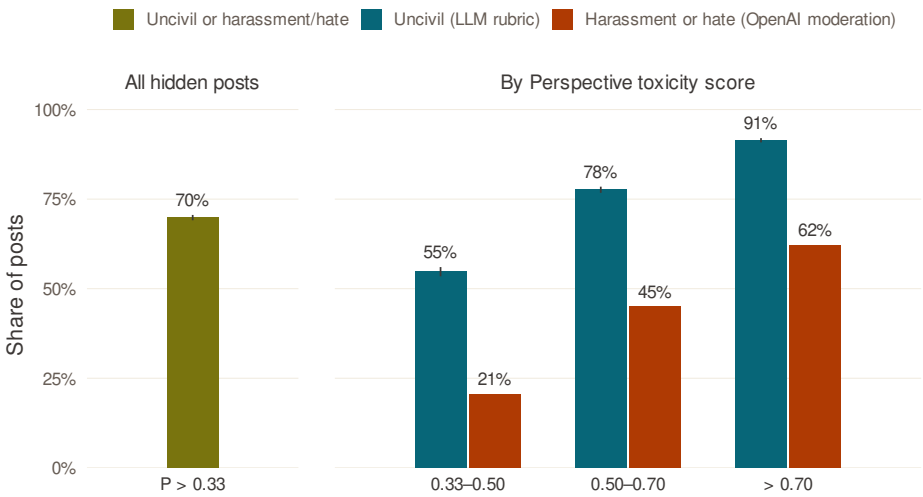


FIGURE 11: ALIGNMENT WITH INCIVILITY AND HATE/HARASSMENT CLASSIFIERS

Notes: This figure reports the share of posts classified as uncivil by a large language model and the share flagged for harassment or hate by OpenAI's moderation model. The analysis was conducted for posts (excluding comments) supplied to participants in the analysis sample during the six-week treatment period. Each post is included only once even if it appeared multiple times. The left panel covers all posts above the hiding threshold (Perspective toxicity score above 0.33) and reports the share that are either classified as uncivil or flagged for harassment or hate. The right panel reports each measure separately by Perspective toxicity score region. Incivility is classified by gpt-6-luna from the post's text alone, using a coding rubric adapted from Coe et al. (2014), on a sample of 47 669 posts drawn at random within strata defined by Perspective toxicity score, Alienation score, and platform, and reweighted to reflect that strata have different sizes. A post is flagged for harassment or hate if OpenAI's moderation model (OpenAI, 2026) finds that it includes harassment or hate, including threatening variants. For this classifier, every post above the hiding threshold is scored (238 749 posts). Bars denote 95% confidence intervals that reflect sampling variation from the stratified design, with a finite-population correction, computed on the logit scale. There are no confidence intervals for the bars reflecting only OpenAI's measure, because all relevant posts were scored.

similar browser extension also finds no evidence that platforms adjust their feeds in response to filtering over similar periods (Beknazar-Yuzbashev et al., 2025). Taken together, it is implausible that our results are explained by an endogenous reaction of the treated platforms' algorithms.

7 Conclusion

This paper reports results from a field experiment with U.S. social media users on the mental-health effects of a content-level intervention that hides toxic content while preserving platform access. Across two recruitment waves, one during an election campaign and one outside of it, we document that filtering out toxicity *worsens* mental health at six weeks. Our difference-in-differences estimates indicate that toxicity hiding increases the index of mental health symptoms by 0.15σ. The intervention increases symptoms of both anxiety and depression according to clinically validated scales – GAD-7 rises by 0.7 points and the PHQ-9 proxy by 1.0 points. Moreover, we observe an increase in the share of participants who

exceed the moderate-symptom threshold by 12.1 pp. A QALY-based calibration indicates a welfare loss of about $158 per treated participant.

Our research complements prior studies documenting negative effects of social media use on well-being by isolating the role of an important component of social media experience – exposure to toxic content. The results indicate that this component does not contribute to the overall negative effect of social media. On the contrary, filtering toxicity increases symptoms of anxiety and depression. The mechanism evidence points to social connections as an explanation. The intervention increases perception of loneliness across participants. Users with a higher share of content from peers and those who use platforms organized around interpersonal exchange rather than topic communities experience stronger effects. A second channel is lower exposure to political toxicity produced by the in-group. We find that most toxic political posts users see are congruent with their own party. The treatment effect is stronger for partisans than independents, and for those with a morally charged attitude to political conflict. Toxic content is bundled with a sense of social presence and a feeling that others share one's strong political views and fears. These may explain why the intervention led to unintended psychological consequences.

Our findings raise a natural question: does this mean content moderation is net harmful and should be abandoned? The answer is clearly no. There are compelling reasons – including offline violence, discrimination, and the well-documented aggregate harms of hateful content – to continue moderating the content on large platforms. The point is rather that the *form* of moderation matters. Our intervention suppressed toxic content above a fixed toxicity threshold, without user input. A less disruptive alternative would be for platforms to flag posts exceeding a toxicity threshold – displaying a warning or a "view anyway" prompt – rather than hiding them outright. Platforms could additionally allow users to set their own toxicity thresholds. Finally, because the cost concentrates where toxic content arrives through peers, platforms could moderate peer-authored content more lightly than content from public figures and institutions – either automatically, or by letting users opt out of moderation of peer content themselves. Taken together, our paper offers an important broad lesson for policymakers – evaluating moderation requires understanding both the content it affects and the social exchanges it changes.

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